USS Pampanito AI Docent — TODO / Not Yet Implemented

Items described in overview.md that are partially or fully unimplemented. Organised by priority.

🔴 High Priority

1. Multilingual Tour Narration Audio

What the doc says: Narration plays for each compartment. **Reality:** The 13 compartment audio files (`01_after_deck.mp3` → `13_forward_deck.mp3`) are English only. No Spanish, French, German, Japanese, or Chinese narration has been recorded or imported.

To do:

- Record or generate narration audio for each of the 6 supported languages
- Name files using the existing convention: `01_after_deck_es.mp3`, `01_after_deck_fr.mp3`, etc.
- The frontend already requests language-suffixed filenames — the files just don't exist yet

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2. Built-in Text-to-Speech for AI Answers

What the doc says: "Text-to-speech delivers answers aloud." **Reality:** There is no server-side `/tts` endpoint. The TTS chain is:

1. Pre-recorded FAQ audio (only 1 question has been recorded — see item 3)
2. ElevenLabs API — requires the visitor to manually enter a personal API key in the Settings panel
3. Browser `SpeechSynthesis` fallback — robotic quality; language support depends on device

In practice, most visitors hear browser SpeechSynthesis — robotic and English-accented regardless of language setting.

To do:

- Add a server-side `/tts` endpoint (OpenAI TTS or ElevenLabs) that takes `text` + `language` and returns MP3 audio
- Remove the requirement for visitors to supply their own API key
- This is the single highest-impact quality improvement available

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3. Pre-Cached FAQ Audio at Scale

What the doc says: "Pre-recorded TTS answers that play instantly without any wait." **Reality:** Only **one** FAQ question (`faq_594`) has pre-recorded audio, in all 6 languages. The remaining ~228 FAQ answers have no audio files — they fall through to ElevenLabs/browser TTS.

To do:

- Identify the top 50–100 most likely questions
- Generate TTS audio for each in all 6 languages using the established naming convention (`faq_{id}_{lang}.mp3`)
- Add audio files to `web/audio/`

🟡 Medium Priority

4. Follow-up Question Suggestions

What the doc says: "The guide proposes related questions to deepen visitor engagement." **Reality:** The `/ask` API does return a `followups` array in every response, but the frontend completely ignores it — nothing is displayed or spoken.

To do:

- After an answer plays, display 2–3 follow-up question buttons in the UI
- Tapping a follow-up should submit it as a new question
- Optionally speak the follow-up options aloud too

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5. Multilingual AI Answer Text

What the doc says: "Answers available in English, Spanish, French, German, Japanese, and Chinese." **Reality:** The extractive synthesis in `api/main.py` always returns English text, because the corpora are in English. Even if TTS could speak other languages, the answer text itself would be English.

To do:

- Either: run answer text through a translation step before returning it (e.g. OpenAI `gpt-4o` translate)
- Or: source/create translated corpora for the target languages
- Neither option is currently implemented

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6. Fallback Audio Files in Multiple Languages

What the doc says: Graceful fallback audio plays when the system can't answer. **Reality:** `nodirectanswer.mp3`, `nothits.mp3`, `emailaddressorphonenumber.mp3`, and `wewillgetbacktoyou.mp3` are English only.

To do:

- Record/generate the four fallback audio clips in each supported language
- Update the frontend to select the language-appropriate fallback file

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🟢 Roadmap (Acknowledged in doc as future work)

7. LLM-Backed Answer Synthesis

Status: Code stub exists (`USE_LLM` flag in `api/main.py`) but the GPT-4o synthesis path has not been implemented or tested. **Blocks on:** OpenAI API budget/funding.

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8. Pre-Cached TTS for Top 50 Questions

Status: Listed as near-term roadmap. Depends on item 2 (server-side TTS) being completed first so audio can be generated at scale.

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9. Expanded Corpus Coverage

Status: Not started.

- Crew oral history transcripts
- WW2 patrol log excerpts
- Post-war interviews
- USS Pampanito-specific technical documents

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10. QR Code Self-Check-In / Analytics

Status: Not started. No analytics instrumentation exists.

- Track which compartments get the most questions
- Track which questions are most frequently asked
- Track language distribution

📋 Documentation Corrections Needed

The overview in `overview.md` contains some inaccuracies introduced before the current architecture was confirmed:

Section	What doc says	Reality
Section 6.2, API table	`POST /tts` endpoint exists	No `/tts` endpoint — TTS is entirely client-side
Section 6.2, Python stack	"OpenAI SDK (TTS only)"	OpenAI SDK is not used at all currently (Whisper removed; no TTS endpoint). ElevenLabs is used client-side if the user provides a key.
Section 5, Features	"Spoken answers" implies reliable TTS	In practice: robotic browser SpeechSynthesis unless visitor enters ElevenLabs key
Section 5, Features	"Multilingual audio"	Tour narration is English only; subtitles/translated answers not implemented

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Last updated: March 2026